

$$\frac{pV}{T} = \frac{pV}{T} \rightarrow \frac{0.2 * 5}{500} = \frac{0.4 * 1.25}{T} \rightarrow T = 250 K$$

$$R \text{ of air} = 0.2870 \text{ kJ/kg} * K$$

$$5 \bar{5} 500 \text{ kPa}$$

$$m = \frac{pV}{RT} \rightarrow m = \frac{500 * 0.2}{0.287 * 500} = 0.6969 \text{ kg}$$

$$\text{at } 500 K, u_1 = 359.49 \text{ kJ/kg}$$

$$\text{at } 250 K, u_2 = 178.28 \text{ kJ/kg}$$

$$Q = W + \Delta U = 0 + m(u_2 - u_1) = \frac{0.2}{0.287} * (178.28 - 359.49) = -126.28 \text{ kJ}$$